

Near Field Communication For Secure Applications

If you find someone using a mobile phone to make a payment at the cash counter, or see a policeman charging penalty for a traffic violation through his handset, don't get surprised. NFC-enabled mobile phones are all set to become popular for applications that need security and ease of operation



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It is an invisible part of our lives, yet wireless short-range communication is emerging as one of the most important technologies for intuitive, simple and safe communication. The new short-range radio frequency technology, near-field communication (NFC), enables exchange of data between devices held in proximity of each other. Using NFC, one can 'pick' information from the environment,

as per the NFC forum, co-founded by NXP, Sony and Nokia, to promote the NFC technology.

"NFC can be defined as a combination of already existing technologies. It is compatible with radio frequency identification (RFID) standards, based on ISO 14443A and 14443B. It also complies with the Smart Card technology for contact-based systems defined in ISO 7816," says Kandasamy S., senior manager-portfolio, Sasken.

According to Kandasamy, NFC

itself has been standardised as a technology in ISO 18092, which includes the Felica standards of Japan. Besides, the unification of various standards allows NFC to be used in all major geographies.

"Nxp India has conceptualised, promoted and implemented the innovative idea of NFC phone/terminal and smart card as a system for financial inclusion," says Ashok Chandak, senior director-global sales and marketing, NXP Semiconductors. The company is involved in multiple pilot projects on financial inclusion like ALW, Integra, C-Sam/NIC, etc.

Short-range wireless technologies

Wireless technologies have brought revolutionary changes not only in gadgets but also the way we carry out various tasks. As every product or application has its own unique requirements, it is a tedious job for an application/product designer to choose the right technology from a plethora of evolving standards and specifications in new technologies. To choose a technology from the existing pool, common criteria are: operating range, interoperability, security, power requirement, data rate and cost.

There are many short-range wireless technologies like Bluetooth, NFC, Zigbee, IR, Wireless USB and Z-Wave. All these work on different operating frequencies and offer different data rates. Of these, Bluetooth is one of the widely-accepted technologies, as it has replaced cables and the need for